Northern Lights Deep Learning Conference Template

Anonymous Full Paper
Submission 42

Abstract

Deep learning is an emerging subfield in machine learning that has in recent years achieved state-of-the-art performance in image classification, object detection, segmentation, time series prediction and speech recognition to name a few. This conference will gather researchers both on a national and international level to exchange ideas, encourage collaborations and present cutting-edge research.

1 Introduction

The Northern Lights Deep Learning Conference will take place in Tromsø, Norway. Being 350 km north of the arctic circle, Tromsø is the northernmost city in the world with a population above 75,000. As Tromsø is located in the middle of the Northern Lights Oval, the area with highest probability of observing northern lights, there is always a good chance to see them. In general Tromsø has a mild climate for such a northerly destination because of its seaside location and the warming effect of the Gulf stream. Tromsø is easily accessible from other Norwegian cities and from abroad. More information about Tromsø is available at Visit Tromsø [1].

2 Submission Guidelines

We invite participants to submit their work in PDF format using our LaTeXtemplate. We do not provide any Word template and we discourage authors from using Word for their submission. The submissions can be either (1) a 2-page extended abstract (non-archival), using the `abstract` option, or (2) a 5 to 8 pages full paper (included in proceedings) in the area of Deep Learning, using the `fullpaper` option. The references do not count towards the page limit. The submissions can have an appendix with unlimited number of pages with supplementary material and information that do not fit in the main paper. However, **the reviewers are not obliged to read the appendix**—thus, all your main results and points must be maid in the main paper. The submissions can be submitted by following the instructions on the conference website.[1] Submissions must be in pdf format (two column, font size 10) using the official NLDL template using the `review` option, and must include the OpenReview ID (obtained after creating the submission) using the macro `\paperID{XX}`, where `XX` is the submission's number. All full paper submissions will undergo double-blind peer review. It will be up to the authors to ensure the proper anonymization of their paper (see Section 3). In summary, do not include any names or affiliations, and refer to your own past work in the third-person. **Full paper submissions that violate any of these restrictions will be desk-rejected without going further review.** Accepted contributions will be accepted as contributing talks or poster presentations. The abstracts will be single-blind peer review (i.e., the authors identity is known but the reviewers are anonymous).

[1] https://nldl.org

3 Anonymized Submission

For full paper submissions, to maintain a fair and impartial review process, we require anonymized submissions. This means that you must carefully follow these guidelines to ensure that your manuscript conceals your identity during the peer review process.

Personal information. Remove all author names and any personal identifying information from the manuscript, such as author bios or acknowledgments. (Note that this does not mean that you cannot cite your previous work. See below for details.) Similarly, do not include the names of your affiliations or institutions in the manuscript. Note that this template **defaults to the anonymized version** (`review`) that automatically removes authors' names and institutions.

Self-References. While it is acceptable to cite your own previous work, do so in the third person to maintain anonymity. For example, write, "Smith et al. [1] explored this concept," instead of "In our previous work [1], we explored this concept." **Do not remove the name of the authors in any of included references.** This information is not considered personal information since it will be considered as any other reference.

4 Camera Ready

Use `final` in the options of the class to produce the camera ready version. This will automatically remove the line numbers and will de-anonymized the submission. For the camera-ready version, the main paper is limited to 8 pages. Authors must

address the reviewers' comments within this limit. However, the appendix is still unlimited. If revisions cause the paper to exceed the page limit, authors should compress the content in the main paper and relocate additional results to the appendix in order to adequately address the comments. The camera ready can have an acknowledgment section that doesn't count towards the page limit. As with the review version, the references do not count towards the page limit.

You have to set the `\vol{V}` macro at the beginning of the template with the volume that was provided to you. Note that **you shouldn't submit the final version without the volume number**.

5 Style Guidelines

1. **Typeface.** Do not change the typeface of the template by adding other packages that change it. For instance, you **should not** use the packages `palatino`, `fourier`, `bookman`, among many others.

2. **Paper identification.** The review version of the paper (set using `review` option in the document class) must include the OpenReview ID (obtained after creating the submission) using the macro `\paperID{XX}`, where `XX` is the submission's number.

3. **Floating elements.** Floats in a paper allow for flexible placement of figures, tables, and other similar objects. When including a figure or table, use the `\begin{figure}[tb]` or the `\begin{table}[tb]` environments to automatically create a float. These environments take an optional location parameter. You must avoid forcing the floats in place with `!hH`. Instead, you should let LaTeX determine the optimal position for your figure or table based on the overall layout of the page, aiming to maintain a clean, coherent layout of the text and figures by using `tb` instead. The locations `t` and `b` suggest to LaTeX that it should place the figure or table at the top or bottom of the page if possible, but will not force it to do so.

   Figure captions should be placed at the bottom while table captions at the top to improve the document's readability. Sample usage is below, with the examples displayed in Fig. 1 and Table 1.

```
\begin{figure}[tb]
  \centering
  \includegraphics[width=\linewidth]{
      example-image-a}
  \caption{This is a sample image.}
  \label{fig:sample}
\end{figure}
```

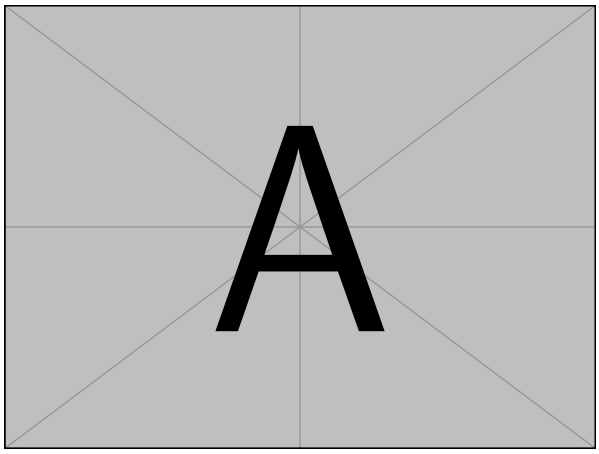

Figure 1. This is a sample image.

Table 1. This is a sample table.

Column1	Column2	Column3
1	2	3

```
\begin{table}[tb]
  \centering
  \caption{This is a sample table.}
  \label{tab:sample}
  \begin{tabular}{lcr}
    \toprule
    Column1 & Column2 & Column3 \\
    \midrule
    1 & 2 & 3 \\
    \bottomrule
  \end{tabular}
\end{table}
```

You must avoid having floating elements beyond the column margins. In case you have floating elements (tables or figures) that go outside of the column margins, you can resize them to fit the column with with `\resizebox{\linewidth}{!}{<float>}`. For instance, if you have a table wider than the column, as in Table 2, you can adjust the content with the `\resizebox` macro, as in Table 3.

```
\begin{table}[tb]
  \centering
  \caption{This is a sample table
      with the wrong width since it
      goes outside of the margins. \
      textbf{Do not do this!}}
  \label{tab:wrong-width}
  \begin{tabular}{lcrrr}
    \toprule
    Column1 & Column2 & Column3 &
        Column4 & Col5 \\
    \midrule
    1 & 2 & 3 & 4 & 5 \\
    \bottomrule
  \end{tabular}
\end{table}
```

Table 2. This is a sample table with the wrong width since it goes outside of the margins. **Do not do this!**

Column1	Column2	Column3	Column4	Col5
1	2	3	4	5

Table 3. This is the correct way of setting Table 2.

Column1	Column2	Column3	Column4	Col5
1	2	3	4	5

```
\begin{table}[tb]
  \centering
  \caption{This is the correct way of
      setting Table~\ref{tab:wrong-
     width}.}
  \label{tab:fixed-width}
  \resizebox{\linewidth}{!}{%
    \begin{tabular}{lcrrr}
      \toprule
      Column1 & Column2 & Column3 &
          Column4 & Col5 \\
      \midrule
      1 & 2 & 3 & 4 & 5 \\
      \bottomrule
  \end{tabular}}
\end{table}
```

Note that you must resize the content and not the float. That is, you must resize `tabular` environment or the `\includegraphics` macro instead of the `figure` or `table` environments.

4. **Mathematics.** Mathematical expressions can be inserted into the text either inline or as standalone entities. For inline mathematical expressions, you can use either the `$$...$$` or `\(...\)` delimiters. However, the `\(...\)` notation is preferred as it offers better error handling. If there are any syntax errors within the `\(...\)` delimiters, LaTeX provides more explicit error messages that help in quickly spotting and correcting the issue.

   For standalone, block equations, use the `\begin{equation}` and `\end{equation}` macros. This also automatically numbers the equations. It is recommended that all mathematical formulas or equations be numbered, whether or not they are explicitly referenced in your text. The equation numbers will assist other researchers who may wish to refer to specific parts of your work in their own publications.

   Variables in mathematical expressions should ideally be represented by a single letter, both for subscripts and superscripts. In cases where more context or clarity is required by including words or multiple letters, you should escape the word using `\mathit`, `\text`, or `\mathrm`. For example:

$$\mathbf{w}^{(\text{new})} = \mathbf{w}^{(\text{old})} - \alpha \frac{\partial \mathcal{L}(\mathbf{w}^{(\text{old})})}{\partial \mathbf{w}}. \tag{1}$$

   But you should prefer to use $\mathbf{w}^t$ instead of the word based superscripts. The mathematics in the manuscript should flow within the language and should not appear as standalone pieces. You are recommended to review the work of Knuth et al. [2].

5. **Code.** For inline code references, use the `\texttt`, `\verb`, or `\lstinline` commands. Larger code blocks can be included using the `verbatim` environment. If pseudocode or more sophisticated code formatting is needed, use the `algorithm2e` or `listings` packages.

   Code blocks that follow the text can appear in between the paragraphs. For larger pieces that will be referred to in the text, it is better to use an `algorithm` environment and use it as a float element (similar to the figures and tables).

6. **Citations.** This style enables textual and parenthetical citations `\citet` and `\citep` through the `natbib=true` option in the `biblatex` package. This style also provides a possessive or genitive citation macro `\citepos` and `\Citepos` that behaves as the textual citations with the addition of the possessive marker at the end.

   Note that citations are not nouns, and you have to use them tied to a noun or a noun-phrase when used as parenthetical citations. With parenthetical citations, you should use non-breakable spaces (`~`) between the word and the `\cite` macro (see the usage throughout this document for examples). E.g., you should write "We refer to some website [1]." instead of "We refer to [1]." Similarly, you can rely on the textual form of the citation instead.

7. **References.** This style uses the BibLaTeX package since it offers a modern, flexible, and convenient way to handle bibliographic data. Ensure to add your bibliographic database (`.bib` file) within the preamble as `\addbibresource{mybibliography.bib}`. To display the bibliography, use the `\printbibliography` command to print it out.

   In this style, we load the `biblatex` package with the `biber` backend. You must ensure the right chain of compilation: `LaTeX => Biber => LaTeX` (twice). Many modern LaTeX editors can streamline this process, match your editor preferences accordingly.

8. **Abbreviations.** This style provides a set of the most common academic abbreviations that

take care of correct punctuation after them. The macros available are: `\eg`, `\Eg`, `\ie`, `\Ie`, `\cf`, `\Cf`, `\etc`, `\vs`, `\wrt`, `\dof`, `\iid`, `\wolog`, and `\etal`.

6 Conclusion

We look forward to see you all in Tromsø for an engaging and productive conference!

Acknowledgments

You can acknowledge whomever you seem necessary here, or cite sources of funding here. Do not use the footnotes in the first page. You should add this section **only in the camera ready version** of the paper. This section doesn't count towards the page limit for the camera ready.

References

[1] V. Tromsø. *Official travel guide to Tromsø*. www.visittromso.no/en. [Online; accessed 23-October-2017]. 2017.

[2] D. E. Knuth, T. Larrabee, and P. M. Roberts. *Mathematical writing*. 14. Cambridge University Press, 1989.

A Example of the Appendix

The purpose of this appendix is to provide additional explanations and clarifications for certain aspects of the paper. While these explanations can enhance the understanding of the proposed method, they are not considered essential to the main content of the paper. Reviewers are not required to read this appendix, as the main information necessary for evaluating the paper is contained within the main paper itself.

The information provided in this appendix includes detailed descriptions of figures, additional analysis, and further discussions related to the proposed method. It aims to provide supplementary insights. For example, Fig. A.1 must provide additional information, but it shouldn't be needed to evaluate the paper. Similarly, the authors shouldn't expect the reviewers to read the appendix to fully understand the paper.

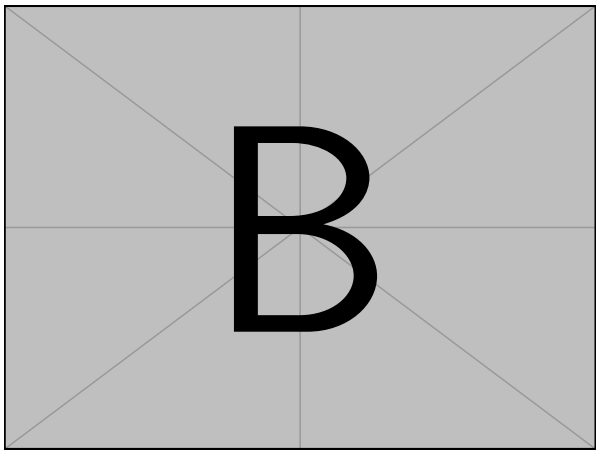

Figure A.1. This is a sample image that the reviewers are not forced to read or review.